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Scleritis and Development of Immune-Mediated Disease: A Retrospective Chart Review

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ABSTRACT. ***Objective.*** Scleritis may be idiopathic or caused by trauma, infections, or an immune-mediated disease (IMD). Our study aimed to understand the relationship between scleritis and IMD, including presenting characteristics, serologies, and treatment course. Understanding these associations may allow clinicians to risk-stratify patients and predict their clinical and treatment course.

Methods. We conducted a retrospective chart review of 341 patients with scleritis seen at a tertiary care center between January 1, 2005, and December 31, 2020. Demographics, scleritis characteristics, treatment response, recurrence, and serologic data were compared among patients with idiopathic and IMD-associated scleritis.

Results. Among patients with scleritis seen, 145 patients (43%) had an associated IMD, most commonly rheumatoid arthritis (RA; 39%), vasculitis (21%), or inflammatory bowel disease (14%). In most cases, the IMD diagnosis predated the scleritis presentation (63%), though vasculitis cases were more likely to develop during or after scleritis episodes. There were no significant differences in demographics or treatment failures among patients with scleritis with and without associated IMDs. Patients with IMDs were more likely to have a recurrence of scleritis (62% vs 49%, $P = 0.02$).

Conclusion. At our ophthalmology center, 43% of patients with scleritis had an associated IMD, and most patients with an IMD were symptomatic from this disease prior to scleritis presentation. RA was the most commonly associated condition and typically predated the scleritis, whereas vasculitis was more likely diagnosed during or after the scleritis episode. Scleritis among patients with IMD is more likely to recur compared to scleritis that is idiopathic.

Key Indexing Terms: immune system diseases, rheumatoid arthritis, scleritis, vasculitis

Scleritis is inflammation of the scleral tissue and may be idiopathic or caused by surgery, infection, or an underlying immune-mediated disease (IMD).[1] Diagnosis relies on symptoms of pain, response to topical phenylephrine, and slit lamp examination. The majority of scleritis presentations are anterior, or proximal to the rectus muscle's limbus insertion point, as opposed to posterior, and can be nodular, diffuse, or necrotizing.[2] The sclera is vulnerable to inflammation because of poor perfusion and drainage and because of the similarity in its composition to other connective tissue, such as joints.[3,4]

Scleritis is associated with an IMD in approximately 22% to 46% of cases.[5-10] In Asia, the association is slightly lower at 13% to 27%, and is notably low in India (5-7%), likely due to a higher proportion of infectious etiology.[11,12] The relative prevalence of each IMD associated with scleritis varies by study; however, rheumatoid arthritis (RA) is typically the most commonly associated IMD (3-29%). Vasculitis (particularly granulomatosis with polyangiitis), inflammatory bowel disease (IBD), spondyloarthritides (SpA), relapsing polychondritis, and systemic lupus erythematosus (SLE) are also known to be associated with scleritis.[5-9,11-24] The associations between scleritis and IMDs may be changing, as therapies for RA and other diseases have become more effective. Scleritis may be treated with nonsteroidal anti-inflammatory drugs (NSAIDs) as well as immunosuppressants, including local and systemic corticosteroids (CS), methotrexate (MTX), azathioprine, mycophenolate, cyclophosphamide, and various biologics.[6,7,10,14,24,25]

In most cases (79-84%), the IMD diagnosis predates the

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development of scleritis.[1,6,20,21,26,27] A minority of patients (7-18%) have been shown to develop systemic IMD after scleritis diagnosis over a range of 2 to 6 years.[1,6,20,21,26,27] Patients with IMD-associated scleritis have been shown to be more likely to have positive serologies, including rheumatoid factor (RF), antinuclear antibody (ANA), and antineutrophil cytoplasmic antibody (ANCA).[14,28] Patients without IMD at scleritis diagnosis who have positive RF and ANCA have been shown to be more likely to develop RA and ANCA-associated vasculitis (AAV), respectively.[29]

Although it is understood that most IMDs develop before scleritis, the purpose of our study was to describe IMD-associated scleritis, including the natural history, treatment response, recurrence of symptoms, and serologic data, all of which have not been well described in literature. Understanding the natural history of IMD-associated scleritis may help inform treatment or screening practices.

METHODS

Patient criteria. Patients were included if they were diagnosed with scleritis at the Oregon Health & Science University (OHSU) Casey Eye Institute Uveitis Clinic (a subspecialty clinic with expertise in the diagnosis and treatment of inflammatory eye diseases) between January 1, 2005, and December 31, 2020, and were aged > 18 years at the time of their presentation. Patients were identified by clinical coding using the International Classification of Diseases, 10th revision, code H15.0 for scleritis. Patients were excluded if they were seen for < 2 visits at this clinic, or if the diagnosis was equivocal. Patients who were diagnosed with medication-induced scleritis, infectious scleritis, and trauma-induced scleritis were included in the initial dataset but excluded from subsequent analyses, which focused on comparing idiopathic cases with those associated with IMDs.

Data collection. Institutional review board approval was obtained from OHSU (study 11326) to review patient electronic medical records (EMRs). Patient charts were reviewed to obtain demographic data, scleritis characteristics, specific treatments and duration, therapeutic success or failure, prior history of IMDs, subsequent IMDs diagnosed during or after the scleritis episode and time to diagnosis, and serologic data. A treatment was considered to have failed if, at a subsequent follow-up visit, the treating ophthalmologist felt the clinical symptoms and/or examination were inadequately controlled on the prescribed therapy, and the patient required a step-up in therapy. This follow-up visit typically occurred within 2 weeks of the start of a therapy. Treatment failure for MTX was defined as requiring step-up therapy in addition to or instead of MTX, typically assessed after at least 4 weeks of treatment. IMD diagnoses were obtained from the medical history and chart review. There was no time limit of chart follow-up to IMD diagnosis; patients' charts were reviewed up to the last available data to determine whether an IMD had been diagnosed.

IMD definition. An IMD diagnosis was temporally categorized as occurring (1) before scleritis if the patient had been diagnosed prior to presenting for scleritis; (2) during the scleritis episode if the IMD was diagnosed within 1 month before or after scleritis diagnosis; or (3) after the scleritis episode if the IMD was diagnosed > 1 month after the scleritis presentation. Within these categories, serologies and the types of IMD developed were compared. IMDs that developed were described as events, as 14 patients had > 1 IMD. Serologic data were collected by chart review and could have been drawn at any time in the patient's history. Serologic data for IMD diagnoses predating the scleritis diagnosis (eg, serologies of a patient with RA with a diagnosis predating scleritis) were recorded if available.

Data analysis. Data analysis was conducted in Microsoft Excel and R 4.3.0 (R Core Team). Demographics (age, sex, ethnicity, and race), scleritis characteristics, and treatment failures were compared between patients with idiopathic scleritis and those with an associated IMD using chi-square analysis. Timing of the IMD diagnosis in relation to scleritis was also compared using chi-square analysis. Odds ratios (ORs) were calculated to compare the presence of certain serologies in patients with new IMD diagnoses during or after scleritis with idiopathic scleritis cases. Significance was determined based on an adjusted alpha value of < 0.03, calculated using a false discovery rate of 5%.

RESULTS

We reviewed 341 patients with scleritis in our study. Among patients with scleritis, 178 (52%) patient cases were idiopathic, 145 (43%) were IMD-associated, 12 (4%) were caused by infection, 5 (1%) were related to trauma or recent surgery, and 1 (< 1%) was related to medication use. We focused our subsequent analysis on the 323 idiopathic and IMD-associated scleritis cases. Patients had available follow-up EMR data for an average of 9 years, although the range of follow-up was broad (range 0.7-23 yrs, SD 5.2). The follow-up period was similar among patients with idiopathic scleritis (9.1 yrs, SD 5.2) and IMD-associated scleritis (9.0 yrs, SD 5.1). The average age at scleritis presentation was 59.6 (SD 15.8) years, and most patients were women (66%), non-Hispanic (84%), and White (83%; Table 1). There were no significant differences in age, sex, ethnicity, or race among patients with idiopathic or IMD-associated scleritis. Thirty-six patients presented with posterior scleritis; these patients were about twice as likely to have idiopathic disease compared to IMD-associated scleritis (72% vs 28% of posterior cases; $P = 0.01$). Otherwise, there were no significant differences in laterality or type of scleritis among idiopathic vs patients with IMD-associated scleritis.

Compared to those with idiopathic scleritis, patients with IMD-associated scleritis were more likely to be treated with topical CS (55% vs 49%, $P = 0.01$), systemic CS (71% vs 58%, $P = 0.01$), and MTX (46% vs 21%, $P < 0.01$), and were less likely to be treated with NSAIDs (46% vs 57%, $P < 0.01$) or locally injected CS (8% vs 13%, $P = 0.01$). There were no differences observed for treatment failures for NSAID, injected CS, systemic CS, or MTX therapy between those with idiopathic and IMD-associated scleritis. Fifty-four percent of patients had a documented recurrence of their scleritis. Patients with IMD-associated scleritis were more likely to have recurrence of their scleritis compared with idiopathic cases (62% vs 49%, $P = 0.02$; Table 1).

Among patients with IMD-associated scleritis, the most commonly associated conditions were RA (n = 57, 39%), vasculitis (n = 31, 21%), and IBD (n = 21, 14%; Table 2). Among the 95 patients who had an IMD diagnosis before developing scleritis, 45% had RA. Among the 57 diagnosed with RA, patients were more likely to have been diagnosed with RA before scleritis compared to during or after scleritis (74% vs 26%, $P = 0.01$). Overall, RA was diagnosed 5.8 years on average before scleritis diagnosis. More patients with IBD were diagnosed before scleritis episodes than during or after their scleritis diagnosis (81% vs 19%, $P = 0.03$), though this finding was not statistically significant when corrected for multiple findings. On average, IBD was diagnosed 9.6 years before scleritis presenta-

Table 1. Characteristics of patients with scleritis.

Category	Characteristic	Total, N = 323	Associated IMD, n = 145	Idiopathic, n = 178	IMD vs Idiopathic, *P*
Age, yrs, mean (SD)		59.6 (15.8)	51.8 (14.7)	59.2 (16.5)	0.63
Sex	Male	109 (34)	51 (35)	58 (33)	0.62
	Female	214 (66)	94 (65)	120 (67)	
Ethnicity	Hispanic	35 (11)	16 (11)	19 (11)	0.89
	Non-Hispanic	270 (84)	120 (83)	150 (84)	
	Unknown	18 (6)	9 (6)	9 (5)	
Race	White	269 (83)	117 (81)	152 (85)	0.52
	Asian	8 (2)	4 (3)	4 (2)	
	Black	15 (5)	7 (5)	8 (4)	
	> 1 race	3 (1)	2 (1)	1 (1)	
	Unknown	28 (9)	15 (10)	13 (7)	
Laterality[a]	Bilateral	140 (43)	69 (48)	71 (40)	0.27
	Unilateral	136 (42)	76 (52)	60 (34)	
Location[a]	Anterior	250 (77)	125 (86)	125 (70)	0.06
	Posterior	36 (11)	10 (7)	26 (15)	**0.01**
	Necrotizing	12 (4)	7 (5)	5 (3)	0.45
	Unknown	6 (2)	3 (2)	3 (2)	
Subtype[a]	Diffuse	258 (80)	120 (83)	138 (78)	0.4
	Nodular	50 (15)	20 (14)	30 (17)	
	Unknown	15 (5)	5 (3)	10 (6)	
Treatment[a]	NSAID	167 (52)	66 (46)	101 (57)	**< 0.01**
	Failure[b]	74 (23)	32 (48)	42 (42)	0.38
	Topical CS	168 (52)	80 (55)	88 (49)	**0.01**
	Failure[b]	51 (16)	28 (35)	23 (26)	0.21
	Injected CS	34 (11)	11 (8)	23 (13)	**0.01**
	Failure[b]	4 (1)	1 (9)	3 (13)	0.74
	Systemic CS	206 (64)	103 (71)	103 (58)	**0.01**
	Failure[b]	51 (16)	20 (19)	31 (30)	0.08
	MTX	104 (32)	67 (46)	37 (21)	**< 0.01**
	Failure[b]	37 (11)	23 (34)	14 (38)	0.81
Recurrence	Yes	183 (54)	90 (62)	87 (49)	**0.02**

Values are expressed as n (%) unless indicated otherwise. Values in bold are statistically significant. [a] Refers to the initial scleritis episode, not recurrence. [b] % in treatment failure rows represents proportion of patients with this scleritis type who failed given therapy. CS: corticosteroid; IMD: immune-mediated disease; MTX: methotrexate; NSAID: nonsteroidal antiinflammatory drug.

tion. Among the 20 patients with scleritis for whom an IMD was diagnosed at the time of scleritis presentation, 9 had vasculitis (8 were AAV and 1 was a non-ANCA small-vessel vasculitis). Unlike in those with RA, patients with vasculitis were more likely to be diagnosed during or after the scleritis diagnosis than preceding the scleritis (71% vs 29%, $P < 0.01$). In the cases for which IMD was diagnosed after scleritis (n = 34), the mean time to diagnosis was 3.9 (SD 3.9) years. The most common IMDs diagnosed > 1 month after scleritis presentation were vasculitis and RA. For vasculitis developing after scleritis (n = 13), the mean time to diagnosis was 2.5 (range 0.2-10.2) years, whereas for RA (n = 10), the mean time to diagnosis was 3.1 (range 0.2-11.2) years.

In addition to the more common conditions discussed, among SLE and sarcoidosis cases, most were diagnosed before scleritis (6 of 7 SLE cases and 5 of 6 sarcoidosis cases). Among 9 SpA cases, 3 were diagnosed after scleritis. Other IMDs diagnosed during or after scleritis included relapsing polychondritis (n = 4), SpA (n = 3), sarcoidosis (n = 1), and SLE (n = 1; Table 2).

An analysis of serologic data focused on patients who did not have a known IMD diagnosis at the time of their presentation (idiopathic vs IMD diagnosed during or after scleritis), as serologic data for those with a preexisting IMD diagnosis was often not repeated (eg, a patient with a diagnosis of RA made elsewhere would typically not have repeat serologies collected; Table 3). Serologic data were recorded at any timepoint. This analysis included 232 patients, 178 of whom remained idiopathic, 20 who were diagnosed with an IMD during the scleritis episode, and 34 who were diagnosed with an IMD after the scleritis episode. Interestingly, among these 232 patients, 191 (82%) had an ANCA serologic test, whereas only 85 (37%) had serologies drawn for RF and/or anticitrullinated protein antibody (ACPA; Table 3). Among those who tested positive for ANCA, 3 remained idiopathic and 16 were diagnosed with vasculitis either during (n = 7) or after (n = 9) presenting with scleritis. ANCA positivity was strongly associated with a diagnosis of an IMD in this group of patients (OR 26.72, 95% CI 7.01-152.59, $P < 0.01$).

Table 2. IMD-associated scleritis cases.

IMD Type	Total IMD, n = 145	IMD Before Scleritis, n = 91	IMD During Scleritis, n = 20	IMD After Scleritis, n = 34		IMD Before vs During or After
	n	n (%)	n (%)	n (%)	Mean Time From Scleritis to IMD, yrs (range)	*P*
RA	57	42 (74)	5 (9)	10 (18)	3.1 (0.2-11.2)	**0.01**
Vasculitis	31	9 (29)	9 (29)	13 (42)	2.5 (0.2-10.2)	**< 0.01**
AAV	27	7 (26)	8 (30)	12 (44)		**< 0.02**
Other vasculitis[a]	4	2 (50)	1 (25)	1 (25)		–
IBD	21	17 (81)	2 (10)	2 (10)	–	0.03
SLE	7	6 (86)	1 (14)	0 (0)	–	–
Sarcoidosis	6	5 (83)	0 (0)	1 (17)	–	–
SpA	9	6 (67)	0 (0)	3 (33)	–	–
Relapsing polychondritis	6	2 (33)	3 (50)	1 (17)	–	–
Other	16	8[b] (53)	2[c] (13)	6[d] (33)	–	–

Values in bold are statistically significant. Percentages in each row were calculated from the total no. of patients with the specific IMD. IMD (no. of patients): [a] small-vessel vasculitis (2), giant cell arteritis (1), Takayasu arteritis (1); [b] antiphospholipid antibody syndrome (2), Behçet disease (1), Still disease (1), myositis (1), Cogan syndrome (1), IgA nephropathy (1), bullous pemphigoid (1); [c] Cogan syndrome (1), APS (1); [d] Mixed connective tissue disease (1), Cogan syndrome (1), MAGIC syndrome (1), IgG4-related disease (1), undifferentiated connective tissue disorder (1), Sjögren syndrome (1). AAV: ANCA-associated vasculitis; ANCA: antineutrophil cytoplasmic antibody; APS: antiphospholipid syndrome; IBD: inflammatory bowel disease; IMD: immune-mediated disease; MAGIC: mouth and genital ulcers with inflamed cartilage; RA: rheumatoid arthritis; SLE: systemic lupus erythematosus; SpA: spondyloarthropathy.

Table 3. Serologies of patients with scleritis without a known IMD at presentation.

Serology	Idiopathic, n = 178	IMD During Scleritis, n = 20	IMD After Scleritis, n = 34	IMD During or After Scleritis vs Idiopathic	
		n (%)		OR (95% CI)	*P*
Total ANCA tested	147 (83)	16 (80)	28 (82)		
ANCA+	3 (2)	7 (44)	9 (32)	–	–
ANCA–	144 (98)	9 (56)	19 (68)	26.72 (7.01-152.59)	**0.01**
Total RF/ACPA tested	59 (33)	9 (45)	17 (50)		
RF+/ACPA+	0	1 (11)	4 (24)	–	–
RF+/ACPA–	3 (5)	2 (22)	4 (24)	7.21 (1.31-51.21)	**0.01**
RF–/ACPA+	0	0	3 (18)	–	–
RF–/ACPA–	45 (76)	6 (67)	6 (35)	0.27 (0.10-0.72)	**0.01**

Values in bold are statistically significant. ACPA: anticitrullinated protein antibody; ANCA: antineutrophil cytoplasmic antibody; IMD: immune-mediated disease; OR: odds ratio; RF: rheumatoid factor.

Among those with available RF or ACPA serologies, a double seropositive result was seen only in patients with RA, and a double seronegative result was more likely to be seen in idiopathic patients (OR 0.27, 95% CI 0.10-0.72, $P = 0.01$; Table 3). The finding of RF+/ACPA– also increased the likelihood of an IMD diagnosis (OR 7.21; 95% CI 1.31-51.21, $P = 0.01$). RF–/ACPA+ was seen only in 3 cases, all of whom had an IMD diagnosis after scleritis (RA, IgG4-related disease, and AAV). Among patients with AAV with scleritis and available serologies (n = 27), 96% were ANCA positive. Neither ANCA positivity nor RF/ACPA positivity was correlated with an increased risk of relapse.

In patients followed for ≥ 5 years, 24/149 (16.1%) patients without IMD developed an IMD compared to 10/63 (15.9%) patients with < 5 years of follow-up.

DISCUSSION

In our review of 341 patients with scleritis, 43% of patients had an associated IMD. Most developed IMDs before scleritis (63%), compared to concurrently (14%) or after scleritis (23%). Although RA was the most common IMD seen in this population, among those who were diagnosed with an IMD during or after the scleritis episode, vasculitides were the most common IMD diagnosis. Compared to patients with idiopathic scleritis, patients with an IMD were more likely to have a recurrence of scleritis (62% vs 49%, $P = 0.02$), less likely to have a posterior scleritis presentation, and more likely to be treated with CS and MTX. Although we initially hypothesized that patients with IMDs would be more likely to fail NSAID therapy, we did not find any significant difference in treatment failure among those with IMDs or idiopathic cases. This may be explained by the higher rate of CS and MTX use in patients with IMD—in other words, it is possible that these patients are treated more aggressively either due to severity or due to existence or suspicion of an underlying IMD.

The most common IMDs in our study resembled those found in other studies, with comparable prevalence.[1,6,8,10-12,16,20,26,30] Compared with other studies, our population had a relatively greater association with IBD (14% compared with 0.5-11%),[1,11,12,16,24,26] though some studies did not comment on IBD prevalence, choosing instead to focus on SpA.[15,30] This may reflect local demographics in the state of Oregon or the fact that this study was done at a tertiary care center. Our finding that IMDs often develop before scleritis is consistent with other studies.[1,30] We are aware of only 2 studies that divided patients into having an IMD before, during or after scleritis; our study found a higher proportion of vasculitis cases among those who had an IMD diagnosed after scleritis (42% of our cases vs 0 and 14% of other studies).[1,21] Our higher vasculitis frequency may be due to local expertise, the high frequency of ANCA serology testing, a longer follow-up period, and/or a larger sample size. Another study has shown a greater tendency for RA to develop after scleritis compared to our population (47% vs our 18%), though a similar cross-sectional case-control paper found the opposite result, at 16%.[26,30] Our study finding that patients are more likely to develop vasculitis after scleritis presentation compared to patients with RA has been demonstrated similarly in a small study of 10 patients with vasculitis and 8 patients with RA with scleritis followed for variable periods < 4 years.[24] The presence of RA after scleritis is likely underreported in our study and in similar studies, since RA serologies are less frequently tested during scleritis episodes and access to prolonged follow-up data is limited. We found that ANCA and RF positivity are more common in patients who develop an IMD, which is in alignment with a prior study showing that patients with idiopathic scleritis with positive serologies were more likely to develop vasculitis and RA.[29] Our finding that ANCA positivity is associated with the presence of IMD-associated disease has also been observed in a previous retrospective study.[28]

The use of NSAIDs, topical prednisolone, systemic prednisone, and MTX in our study population was within the wide ranges of use seen in other studies.[6-8,10,12,15,16,18,20,21,24,27,31,32] Few studies evaluated treatment response,[16,21,27] and 2 studies similarly found no significant differences in treatment failure between patients with and without IMD.[21,27]

The strengths of our study include a large sample size, a high level of expertise in the diagnosis of scleritis, and longitudinal data collected through long-term EMR review. We also included patients with IMD with a broad variety of diagnoses, notably many vasculitis cases as well as patients with rare diseases such as relapsing polychondritis. Weaknesses of this study include variability in drawn serologies, variability in time of establishment and diagnosis of scleritis, variability in time to determination of treatment failure, lack of data related to number of relapses or time to first relapse, and variable follow-up time. Further, there is the potential that IMD cases diagnosed later or outside of the OHSU system may have been missed. Although follow-up time ranged from 0.7 to 23 years, we found similar rates of IMD development in comparing < 5 years of follow-up to ≥ 5 years. We were unable to capture details regarding IMD or scleritis severity, effect of comorbid conditions, IMD-specific treatments, or mortality rates. Our analysis of treatment response was also limited, as ophthalmologists with a high index of suspicion for an IMD may have chosen to treat with more aggressive initial therapy, and thus the results are not unbiased. It is possible that our study did not allow enough time for a treatment to be considered ineffective, though our report reflects the judgment of the treating ophthalmologist when they determined that treatment was inadequate. Though our study population generally reflects racial and ethnic demographics of Oregon, the results may not be generalizable to more ethnically or racially diverse populations. As cases were seen at a tertiary referral center, the prevalence and severity of associated IMD conditions may be higher in our study populations compared to other clinical settings. Notably, a small portion of our study population (patients seen from 2005 to 2006) had been studied in a previous retrospective review at OHSU on serologic data in patients with scleritis.[28] Last, it is difficult to draw conclusions about some rare scleritis presentations or IMD populations, such as necrotizing scleritis and SLE, as these numbers are small; presence of peripheral ulcerative keratitis was also not included. Although we have provided serologic data in this report, we wish to emphasize that no rheumatic disease is diagnosed solely by serology. Consequently, we are probably underestimating the extent of AAV among patients with scleritis. We also find a fecal calprotectin to be a useful screening test for IBD, but this test was obtained too inconsistently for us to comment on its merit among patients with scleritis.

Key results from our study include the finding that IMD diagnoses are common among patients with scleritis and may be diagnosed at variable times before, during, and after scleritis presentation. Second, we found that scleritis relapses were correlated with the diagnosis of an IMD, and that a posterior scleritis presentation negatively correlated with an IMD. Finally, although RA was the most common IMD associated with scleritis, among patients whose IMD was diagnosed during or after the scleritis episode, 41% were diagnosed with vasculitis. These diagnoses are critical, as vasculitides, particularly AAV, can have life-threatening consequences if treatments are delayed. Ophthalmologists and other providers seeing patients with scleritis should be encouraged by these results to evaluate for vasculitis in scleritis cases. Population surveys or controlled studies with more complete serology data would be useful for determining the relationship between serologies, severity of scleritis, recurrence rates, and treatment differences. Future researchers could focus on additional factors that may affect scleritis severity and treatment, with longer follow-up observation, including IMD severity and systemic involvement, presence of peripheral ulcerative keratitis, and use of biologic therapies.

In conclusion, patients with scleritis are likely to have an associated IMD that may develop before, during, or after presentation. RA, vasculitis, and IBD were the most frequently associated disorders in our study population. Patients who develop

IMDs after scleritis are particularly likely to have vasculitis. More studies are needed to explore the implications of associated IMDs on the risk for severe scleritis episodes and expected treatment responses.

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